

SHADOWFOX

CYBER SECURITY INTERNSHIP

INTERNSHIP REPORT

Name:	MANOJ	Domain:	Cyber Security
Program ID:	SF-PID-2026-W4884	Internship Program:	ShadowFox

INTERNSHIP OVERVIEW

The ShadowFox Cyber Security Internship is a hands-on training program designed to build practical, industry-relevant skills through structured, task-based learning across three progressive levels – Beginner, Intermediate, and Advanced. The program emphasizes real-world application of security concepts, including network reconnaissance, traffic analysis, encryption, executable analysis, and penetration testing, enabling interns to develop a practical understanding of offensive and defensive security practices.

OBJECTIVES

- Perform network reconnaissance and enumeration on target systems
- Analyze network traffic to identify communication patterns and security issues
- Conduct security testing using industry-standard tools
- Understand encryption mechanisms and executable file structure analysis
- Perform penetration testing including exploitation and privilege escalation
- Identify vulnerabilities and document findings with security recommendations

WORK COMPLETED

Beginner Level – Completed

- Port scanning of testasp.vulnweb.com
- Directory enumeration of testasp.vulnweb.com
- Login/network traffic analysis using Wireshark

Intermediate Level – Completed

- Solved a VeraCrypt encrypted file challenge
- Analyzed executable entry points using PE Explorer
- Performed a Metasploit reverse-shell lab on a Windows 10 virtual machine
- Conducted a wireless security assessment on an authorized test network

Advanced Level – Completed (TryHackMe – Basic Pentesting)

- Reconnaissance and enumeration of the target machine
- Identification of vulnerabilities on the target
- Initial access and exploitation as documented
- Privilege escalation as documented
- Documentation of findings and security recommendations

TOOLS & TECHNOLOGIES

Nmap, Wireshark, Metasploit, VeraCrypt, PE Explorer, TryHackMe platform, Windows 10 VM, wireless security assessment tools

SKILLS GAINED

Through this internship, practical skills were developed in network scanning and enumeration, packet-level traffic analysis, encrypted data handling, executable structure analysis, exploitation techniques, privilege escalation, wireless security assessment, and professional security documentation and reporting.

OVERALL LEARNING / CONCLUSION

Completing all three levels of the ShadowFox Cyber Security Internship strengthened my practical understanding of cybersecurity fundamentals and applied techniques. The progression from beginner reconnaissance and traffic analysis to intermediate encryption and exploitation labs, and finally to advanced penetration testing on TryHackMe, gave me hands-on exposure to vulnerability assessment, exploitation, privilege escalation, and structured security documentation. This experience has significantly improved my confidence and readiness to work on real-world cybersecurity tasks.